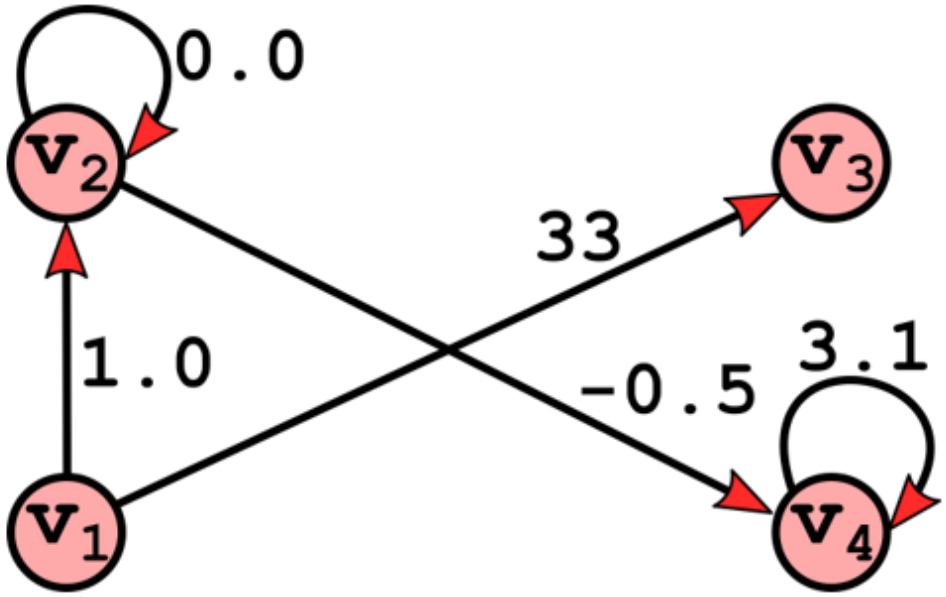


Figure 14.3 A directed weighted graph, with loops. (Edge labels are omitted for the sake of clarity)

We won't bother about multigraphs in this book, so we'll focus on simple graphs, usually without loops.

We can express the previous definitions more formally; given the set of edges E:

- For simple graphs, $E \subseteq \{(x, y) \mid (x, y) \in V2 \land x \neq y\}$
- For simple graphs supporting loops, $E \subseteq V2$

It's also possible to associate a weight to each edge: in this case, the graph is called a *weighted graph* or, equivalently, a *network*, each edge becomes a triplet, and the set of graph's edges becomes:

- For simple graphs, $E \subseteq \{(x, y, w) \mid (x, y) \in V2 \land w \in \mathbb{R} \land x \neq y\}$
- For simple graphs supporting loops, $E \subseteq V2 \times \mathbb{R}$